

Object Oriented Programming

BSIT - Fall 2024

Semester 3

Quiz - 4

Time Allowed: Mins

Total Marks:

1. Multiple Choice Questions:

(8 marks)

1. Consider:

```
int x = 42;  
const int *p = &x;
```

Which of the following is true?

- a) *p = 10; is valid.
 - b) p = &x; is invalid.
 - c) The value pointed to by p cannot be modified through p.
 - d) p must always point to a constant object.
2. What is the primary purpose of a pointer variable?
- a) To store a direct value like an integer or character.
 - b) To store the memory address of another variable.
 - c) To perform arithmetic operations directly.
 - d) To replace array notation.
3. Which of the following statements about NULL and nullptr is correct in modern C++?
- a) NULL and nullptr are interchangeable in all cases.
 - b) nullptr has its own type std::nullptr_t.
 - c) NULL is safer than nullptr because it is strictly typed.
 - d) nullptr can be implicitly converted to any integral type.

4. Consider:

```
int x = 10;  
const int *ptr = &x;
```

Which of the following operations is allowed?

- a) *ptr = 20;
- b) ptr = &y; (where y is another int)
- c) x = 15;
- d) Both b and c

5. Consider:

```
int arr[5] = {1, 2, 3, 4, 5};
int *p = arr;
p++;
cout << *p;
```

- a) It will print 1.
- b) It will print 2.
- c) It will cause a compilation error.
- d) It will cause a runtime error.

6. Consider:

```
int x[] = {10, 20, 30};
int *p1 = x;
int *p2 = &x[2];
cout << p2 - p1;
```

- a) 2
- b) 8 (assuming int is 4 bytes)
- c) 20
- d) 30

7. What will the following print?

```
int arr[5] = {1,2,3,4,5};
int *p1 = arr;
int *p2 = arr + 3;
cout << p2 - p1;
```

- a) 2
- b) 3
- c) Undefined behavior
- d) Depends on compiler

8. Which of the following best explains why incrementing an array name (e.g., arr++) is invalid while incrementing a pointer (e.g., p++) is valid?

- a) Array names are not variables, they are constant pointers.
- b) Array names are lvalues that point to heap storage.
- c) Array names decay into pointers only in function calls.
- d) Array names have static storage duration.

2. Write the output of the following code snippets:

(6 marks)

Code snippet	Output
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```
#include <iostream>
using namespace std;

int main() {
    int nums[] = {5, 12, 8, 21, 6};
    int *p = nums;
    p += 2;
    *p = *p + 1;
    p--;
    *p = (*p) * 2;
    p += 3;
    *p = 100;

    for (int i = 0; i < 5; i++) {
        cout << *(nums+i) << " ";
    }
    return 0;
}
```

3. Write a function `int* findMax(const int *arr, int size)` that takes an integer array and its size as input and returns a pointer to the maximum element in the array. You must **use only pointer notation (no array subscript []). No need to write a main. (6 marks)**

4. A factory has a conveyor belt carrying items (represented as integers in an array). The supervisor wants the order of items reversed before they reach the packing unit.

However, the control system has **some strict limitations**:

- You **cannot** use array indexing (`arr[i]` is forbidden).
- You **cannot** use for or while loops.
- You must solve the problem **using only pointers and recursion**.

Implement the function:

void reverseArray(int *arr, int n);

(10 marks)
